

Pump Flow Compensation and Variable-Gain Sliding Mode Control for Mitigating Temperature-Induced Degradation in Servo Pump-Controlled Systems

Zhiqiang He , Fugang Zhai, Zhuofan Yang , Yangtao Xing , and Xiangdong Kong 

Abstract—Hydraulic-oil temperature variations are a non-negligible factor in servo pump-controlled systems. As temperature rises, viscosity decreases, which induces pump-flow nonlinearities and weakens throttling authority, thereby degrading positioning accuracy and repeatability. To mitigate these effects, this article proposes a temperature-aware composite control strategy that integrates pump-side inner compensation loop with a position-based variable-gain sliding-mode outer robust loop. An inverse-amplification scheme based on online pump-efficiency estimation is coupled with variable-gain sliding mode control: the inner loop compensates pump-flow nonlinearities, while the outer loop suppresses multi-source uncertainties and dynamic disturbances. Together, they provide a two-layer safeguard of “source-level compensation + robust outer-loop regulation.” Comparative tests across multiple operating conditions show that, relative to the baseline proportional-integral-derivative (PID), the proposed strategy improves both repeatability and positioning accuracy by more than 25%. Under nominal and high-speed conditions, it achieves accuracy improvements exceeding 70%. These results demonstrate markedly enhanced high-precision positioning robustness and temperature tolerance under elevated-temperature disturbances and long-term temperature drift.

Index Terms—Inversion amplification method, pump flow compensation (PFC), servo pump-controlled system (SPCS), temperature resistance, variable-gain sliding mode control (VG-SMC).

I. INTRODUCTION

SERVO pump-controlled systems (SPCS) are widely used in construction machinery, computer numerical control (CNC) machine tools, and injection molding machines [1]. In SPCS, a servo motor drives a fixed-displacement pump to realize variable-flow actuation. Compared with conventional variable-displacement pumps, SPCS avoid overflow and throttling losses [1], provide faster dynamic response, and enable higher control accuracy, thereby improving actuator precision and stability [2].

Although SPCS offer high control performance, their motion accuracy and stability remain sensitive to hydraulic oil temperature (HOT) variations [3]. As HOT increases, oil viscosity decreases. This change results in nonlinear deviations in pump flow and weakens the damping of throttling elements [4], which undermines precise and stable motion. Fig. 1 shows the relationship between HOT and motion accuracy over 6000 cycles with the oil cooler operating. HOT rises gradually during repeated motion, and the positioning error increases accordingly, indicating a strong correlation. In addition, the error band widens over time in Fig. 1(d) and follows a trend similar to the HOT rise in Fig. 1(b). These observations suggest that HOT behaves as a slow, long-timescale disturbance that can markedly degrade motion performance. Suppressing HOT-induced nonlinear effects is therefore critical for improving SPCS performance.

To mitigate the adverse effects of HOT variations, existing studies can be broadly grouped into three categories: HOT regulation, error-compensation control, and predictive control.

A. HOT Regulation

Regulating HOT is the most direct way to mitigate temperature-induced nonlinearities. Huang et al. [5] proposed an improved Smith fuzzy-proportional-integral-derivative (PID) scheme for deep-sea applications and reported strong robustness. Li et al. [6] developed a phenomenological pressure-response model that offers high accuracy and an easily invertible

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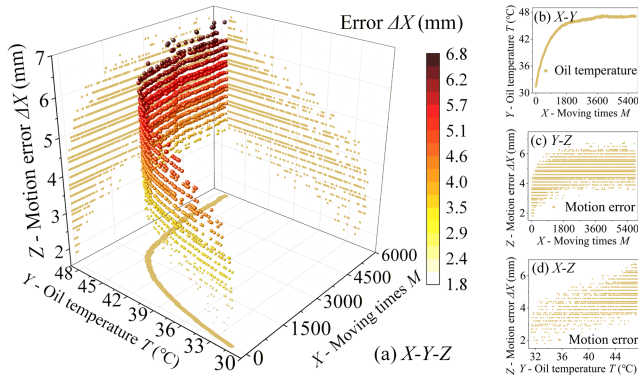


Fig. 1. Scatter plot of hydraulic oil temperature and positioning error during the motion of a SPCS.

form. Chiang et al. [7] designed a PID-based oil-cooler controller with a regulation precision of $\pm 0.2^\circ\text{C}$.

B. Error-Compensation Control

Alternatively, performance can be stabilized by compensating HOT-induced nonlinearities. Li et al. [8] developed a generalized hysteresis model that captures nonlinear behavior and improves tracking accuracy. Shao et al. [9] proposed an adaptive fuzzy strategy to alleviate both steady-state and transient degradation caused by HOT. Zhao et al. [10] integrated a viscosity-based speed-regulation method with PID control, reducing speed fluctuations by more than 30%.

C. Algorithm-Based Predictive Control

Algorithmic prediction has emerged as a third route of work. Ren et al. [11] developed an improved AdaBoost two-layer learner to predict HOT and its categories in tunneling machines. Wang et al. [12] built a BP neural-network model for hydraulic retarders and achieved accurate temperature prediction. Pan et al. [13] proposed a temporal re-attention LSTM predictor with decoupling multivariable, enabling it to handle cross-coupling effects.

Although some progress has been made, three gaps remain as follows.

- 1) **Direct HOT regulation** relies on auxiliary hardware and responds slowly because of thermal inertia [5].
- 2) **Compensation control** is usually evaluated using second-level samples under fast-varying conditions [8], [9]. Their effectiveness under hour-scale, slowly drifting HOT remains insufficiently studied.
- 3) **Data-driven approaches** often adopt shallow models that cannot capture complex nonlinearities. They apply uniform feature weights, which is unsuitable for large time-delay regimes and lack effective temporal-feature extraction, degrading prediction and control [13].

On the other hand, compensating throttling elements, common auxiliary components in SPCS, is more challenging under HOT variations. Pressure dynamics are often characterized using physics-based models [14], [15]. However, the associated flow

mappings are typically highly nonlinear, which complicates controller design and implementation. Polynomial fitting can improve model accuracy with only a few parameters, but the assumption $Q \propto \Delta p$ limits its extension to throttling elements with complex geometries [16]. HOT-induced variations in fluid compressibility are also nonnegligible. Compensation based on the effective bulk modulus has therefore attracted increasing attention [17], [18], [19]. Yet stable parameter adaptation remains difficult because the control-volume parameter enters the pressure-dynamics equation in the denominator. As a result, direct compensation at throttling elements often cannot provide the throttling capacity required in practice.

Given the limitations of component-level regulation, research has shifted toward system-level control. Classical methods, including PID, model-based control, and nonlinear compensation, are structurally simple. However, they rely on predefined models or fixed architectures and therefore adapt poorly to time-varying HOT effects [20], [21]. Adaptive control, robust control, and observer- or disturbance-compensation methods tolerate uncertainties more effectively [20], [22]. Yet under temperature-rise conditions they often become conservative or complex to implement, making it difficult to achieve both dynamic adaptability and high-precision positioning [23], [24]. In contrast, sliding-mode control (SMC) and its variants require less model fidelity and are relatively insensitive to parameter uncertainties and external disturbances [25]. They are thus well suited to SPCS subject to pronounced HOT fluctuations.

However, conventional fixed-gain SMC often suffers from chattering. It also faces a fundamental tradeoff between robustness and steady-state accuracy [26]. These issues have motivated variable-gain SMC (VG-SMC). Existing studies can be broadly grouped into three strands.

- 1) Higher-order SMC is combined with state observers to improve state estimation and alleviate chattering [27].
- 2) Adaptive SMC is integrated with advanced sliding manifolds via state-dependent gain scheduling or control-law reconstruction, including terminal [28], piecewise [29], and fractional-order or super-twisting formulations [30].
- 3) Adaptive SMC is co-designed with complementary mechanisms, such as PID control [29], [31], neural networks [32], or disturbance observers [33], to address channel-specific nonlinearities and multisource uncertainties.

These approaches have been validated in complex systems, including electric drives [28], aircraft [29], underwater platforms [31], and robotic system [34]. Overall, VG-SMC offers clear advantages. Nevertheless, a systematic framework tailored to SPCS remains limited. This gap is most evident for fluid-power-specific composite nonlinearities induced by temperature rise, including pump-source flow nonlinearity, reduced throttling damping, and their coupling with multisource uncertainties.

Motivated by the above limitations in temperature-resilient control and engineering constraints, a temperature-aware composite strategy is proposed that integrates pump-side internal compensation with a position-based VG-SMC outer loop. The main contributions are as follows.

- 1) **Pump-Flow Inversion Compensation**: an inversion amplification scheme based on the pump efficiency η_v is

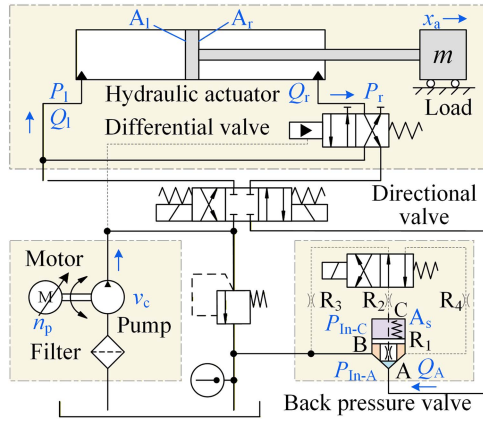


Fig. 2. Diagram of the SPCS.

proposed. The method uses $1/\eta_v$ to amplify temperature-induced efficiency loss into a feedforward correction of the motor-speed command. It is not limited by hardware thermal inertia, and is transferable across systems.

- 2) *VG-SMC Reconstruction for Compound Nonlinearities*: VG-SMC is reformulated with a position-based sliding surface and a variable-gain reaching law, enabling the control gain to adapt to temperature, pressure differential, and the sliding residual. The design improves dynamic adaptability under hour-scale, slowly varying HOT.
- 3) *Composite Architecture*: a composite strategy is developed that integrates source-level compensation with robust outer-loop regulation. The inner loop compensates pump-source nonlinearities, while the outer loop suppresses multisource uncertainties. Simulations and experiments over multiple operating conditions quantify the performance gains and isolate the contribution of each control configuration.

The rest of this article is organized as follows. Section II presents system modeling and the problem statement. Section III presents the control strategy. Section IV reports experimental validation and analysis. Finally, Section V concludes the article.

II. SYSTEM MODELING AND PROBLEM STATEMENT

The operating principle of the SPCS is shown in Fig. 2. A servo motor drives a fixed-displacement pump to directly supply the hydraulic actuator, while a directional valve controls extension and retraction. During extension, a differential valve forms a differential circuit to increase speed. During retraction and deceleration, a back-pressure valve (BPV) provides counterpressure, which enhances braking capability in rapid reciprocating motion. These principles underlie the system model and the subsequent analysis of factors governing motion accuracy and stability.

A. System Modeling

System responsiveness and positioning accuracy are primarily determined by the pump and the BPV. The pump supplies flow to drive actuator motion, whereas the BPV regulates back pressure

and thus affects positioning accuracy. Actuator displacement and velocity, together with pressures at multiple circuit locations, are acquired using built-in sensors.

1) *Gear Pump Modeling*: Leakage dominates pump-flow nonlinearity and can be reflected by the volumetric efficiency. For gear pumps, external leakage is negligible due to sealing, so the efficiency is dominated by internal leakage. Internal leakage mainly occurs through axial leakage, radial leakage, and gear-mesh leakage [35]. Each component is modeled separately.

Axial leakage mainly occurs in the discharge and transition zones of the outer and inner gears. It accounts for about 75%–80% of the total leakage. The axial-leakage flow is given by

$$\Delta q_A = \frac{K_A}{\mu(T)} \cdot \Delta p \quad (1)$$

where Δp is the pressure differential between the high- and low-pressure chambers; $\mu(T)$ is the dynamic viscosity of the hydraulic fluid; and K_A is the viscosity-dependent axial-leakage coefficient of the gear pump.

Radial leakage originates from tip clearances between gear addendums and crescent plates. It accounts for about 15%–20% of the total leakage. The radial-leakage flow is given by

$$\Delta q_R = \frac{K_{R1}}{\mu(T)} \cdot \Delta p - K_{R2} \cdot n_p \quad (2)$$

where K_{R1} represents the viscosity-dependent radial-leakage coefficient; K_{R2} is the viscosity-dependent radial friction-drag leakage coefficient; and n_p is the actual pump speed.

Gear-mesh leakage is negligible. Therefore, the total internal leakage of the gear pump can be approximated as the sum of the axial and radial leakages:

$$\Delta q_p = \Delta q_A + \Delta q_R = \frac{K_A + K_{R1}}{\mu(T)} \cdot \Delta p - K_{R2} \cdot n_p. \quad (3)$$

The outlet-flow equation of the gear pump is given by

$$Q_a = v_c \cdot n_p - \Delta q_p \quad (4)$$

where v_c denotes the constant displacement of the pump.

Therefore, the gear pump flow equation is given by

$$Q_a = (v_c + K_{R2}) \cdot n_p - \frac{K_A + K_{R1}}{\mu(T)} \cdot \Delta p. \quad (5)$$

Because the internal structure of gear pumps is compact, microclearances and leakage paths are difficult to measure directly. A parameter back-calculation identification method based on volumetric-efficiency test data is therefore employed to estimate leakage-related equivalent parameters. Specifically, efficiency tests are conducted over a range of temperature, pressure differential, and speed. The measured data are fitted and visualized to construct the efficiency map $\eta_v(T, \Delta p, n_p)$, as shown in Fig. 3. A mapping between pump performance and model parameters is then established, which enables inverse identification of the key parameters in the theoretical leakage model. As a result, the engineering-oriented flow equation of the internal gear pump is

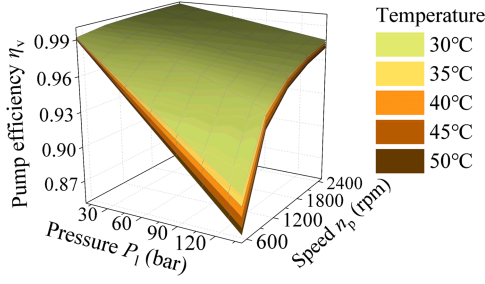


Fig. 3. Gear pump efficiency map under varying pressure and speed conditions.

rewritten as

$$Q_a = v_c n_p \cdot \left[1 - \frac{C_{\text{leak}} \cdot \Delta p}{\mu(T) \cdot v_c n_p} \right] \quad (6)$$

where C_{leak} is a lumped fitted coefficient that aggregates the effects of $\mu(T)$, K_A , K_{R1} , K_{R2} .

2) Back Pressure Valve Modeling: The BPV is installed in the return line of the SPCS, and its characteristics directly determine the braking capability, as shown in Fig. 2. In response to the command signal, the BPV switches between large- and small-throttling states to provide the required parking back pressure. When the spool is open, the force-balance equation is

$$A_s \cdot (P_{\text{In-A}} - P_{\text{In-C}}) = K_s \cdot (z + \Delta z) + C_s \cdot \dot{z} - F_{\text{bias}} \quad (7)$$

where $P_{\text{In-A}}$, $P_{\text{In-C}}$ denote the pressures in chambers A and C, respectively; A_s is the effective area; K_s is the spring stiffness; C_s is the spool viscous-damping coefficient; z is the spool displacement; Δz is the spring precompression; and F_{bias} is the lumped bias load (tolerances, friction, etc.).

Once the spool opens, fluid in chamber A flows through orifices R_1 – R_4 (per the throttling state) and the pilot valve into chambers B and C. The inlet-flow equation at BPV port A is

$$Q_A = Q_{AB} + Q_{AC} + \frac{V_{\text{In-A}}}{\beta_e} \cdot \dot{P}_{\text{In-A}} + A_s \cdot \dot{z}. \quad (8)$$

Using the turbulent square-root orifice model, the pressure-flow relations for each BPV flow path are given by

$$Q_{ij} = R_{ij}(u_v, T) \cdot \sqrt{P_{\text{In-i}} - P_{\text{In-j}}} \quad (9)$$

where Q_{ij} is the flow through passage ij ; $R_{ij}(u_v, T)$ is the opening- and temperature-dependent discharge coefficient; V_i is the chamber volume i ; β_e is the effective bulk modulus; and $P_{\text{In-i}}$ is the pressure in BPV internal chamber i .

The BPV has two operating states: large throttling and small throttling. The orifice radii satisfy $R_1 = R_2 = R_3 > R_4$. Under the small-throttling state, the pilot-circuit flow is given by

$$Q_{AC} = Q_{CB}. \quad (10)$$

The flow-continuity equation at the BPV outlet port B is

$$Q_{AB} = Q_{C3} + \frac{V_{\text{In-B}}}{\beta_e} \cdot \dot{P}_{\text{In-B}} - A_s \cdot \dot{z}. \quad (11)$$

Under the large-throttling state, the BPV pilot flow is

$$Q_{AC} = Q_{C4}. \quad (12)$$

Thus, Q_{AB} is given by

$$Q_{AB} = Q_{C4} + \frac{V_{\text{In-B}}}{\beta_e} \cdot \dot{P}_{\text{In-B}} - A_s \cdot \dot{z}. \quad (13)$$

Accordingly, the inlet-flow equation of the BPV is

$$Q_A =$$

$$\begin{cases} Q_{C3} + Q_{AC} + \frac{V_{\text{In-B}}}{\beta_e} \cdot \dot{P}_{\text{In-B}} + \frac{V_{\text{In-A}}}{\beta_e} \cdot \dot{P}_{\text{In-A}}, u_v = \text{small} \\ Q_{C4} + Q_{AC} + \frac{V_{\text{In-B}}}{\beta_e} \cdot \dot{P}_{\text{In-B}} + \frac{V_{\text{In-A}}}{\beta_e} \cdot \dot{P}_{\text{In-A}}, u_v = \text{large} \end{cases} \quad (14)$$

3) Displacement-Based System Modeling: Based on the system principle in Fig. 2, a displacement-based model of the actuator in the SPCS is established. By the flow-continuity relations, the flows in actuator are given by

$$\begin{cases} Q_l = A_l \cdot \dot{x}_a + C_t (P_l - P_r) + \frac{V_l}{\beta_e} \cdot \dot{P}_l \\ Q_r = A_r \cdot \dot{x}_a - C_t (P_r - P_l) - \frac{V_r}{\beta_e} \cdot \dot{P}_r \end{cases} \quad (15)$$

where C_t is the internal leakage coefficient; x_a is the actuator displacement; P_l and P_r are the chamber pressures; V_l and V_r are the effective volumes; and A_l and A_r are the effective areas.

The force balance equation of the actuator is formulated as

$$P_l \cdot A_l - P_r \cdot A_r + F_d = m\ddot{x}_a + b\dot{x}_a + kx_a \quad (16)$$

where m is the equivalent mass of the actuator and external load; b is the viscous damping coefficient; k is the equivalent stiffness of the actuator-load coupling; and F_d is an external disturbance force (assumed constant over the interval).

Let $A_e \approx A_l \approx A_r$ and $V_t = V_l + V_r$. Subtracting the two equations in (15) and combining them with (16) yields

$$\begin{cases} Q_a \cdot \beta_e = V_l \cdot \dot{P}_l + V_r \cdot \dot{P}_r \\ A_e \cdot (P_l - P_r) + F_d = m\ddot{x}_a + b\dot{x}_a + kx_a \end{cases} \quad (17)$$

To solve the coupled differential equations, we take the Laplace transform of (17), which yields

$$\begin{cases} Q_a(s) \cdot \beta_e = s \cdot P_l(s) V_l(s) + s \cdot P_r(s) V_r(s) \\ A_e \cdot [P_l(s) - P_r(s)] + F_d(s) = X_a(s) \cdot [ms^2 + bs + k] \end{cases} \quad (18)$$

With pump flow Q_a as the primary input, the actuator displacement transfer function is obtained as

$$X_a(s) =$$

$$\frac{A_e \cdot [s \cdot \overbrace{P_l(s)}^{\text{variableterm}} \cdot \overbrace{V_t(s) - \beta_e}^{\text{variableterm}} \cdot \overbrace{Q_a(s)}^{\text{variableterm}}] + s \cdot \overbrace{V_r(s)}^{\text{variableterm}} \cdot F_d(s)}{s \cdot V_r(s) \cdot (ms^2 + bs + k)}. \quad (19)$$

B. Analysis of Influencing Factors

According to the displacement transfer model in (19), the key variables include the pump flow Q_a , the rodless-chamber pressure P_l , and the rod-end chamber volume V_r .

TABLE I
RELEVANT PARAMETERS OF THE SPCS

Item	Value
Hydraulic actuator stroke	$S_{\max} = 456$ mm
Cavity diameter of actuator	90 mm
Piston diameter	55 mm
Rated pressure of the system	175 bar
Rated speed of motors	2800 rpm
Gear pump displacement	63 cc/r
Oil bulk modulus	1000 MPa
Actuator motion speed	60 %

*The motion speed is set as a percentage of the rated speed of motors.

- 1) With a fixed command, variations in the flow Q_a are mainly caused by HOT-induced changes in volumetric efficiency.
- 2) The rodless-chamber pressure P_l is primarily determined by the BPV. During retraction, fluid from the rod-end chamber returns to the tank through the directional valve and the BPV. The pressure losses in the directional valve and piping are small compared with the BPV drop. The back pressure is therefore governed mainly by oil viscosity and the return-flow rate.
- 3) The rod-end chamber volume V_r varies with actuator position. It is a system state variable, rather than an external factor that drives performance variations.

In summary, SPCS motion is sensitive to HOT variations. The temperature effect is mainly mediated by viscosity changes, which modify back pressure and leakage flow and thereby affect system performance. Accordingly, these factors are treated as independent variables in the subsequent analysis. Based on the trends in Fig. 1, a temperature range of 30 °C–50 °C is selected for the study.

To quantify the problem, two metrics are defined. Positioning accuracy measures how close the actuator endpoint position is to the desired position at the end of each motion. Repeatability measures the consistency of the final position over repeated motions under identical conditions. Smaller values indicate higher accuracy and stability.

An electromechanical-hydraulic co-simulation model of the SPCS is developed with explicit consideration of HOT variations. The modeling treatment of nonlinearities (e.g., friction, inertial coupling, and multidisciplinary coupling) follows [36]. Key system and model parameters are given in Table I. Model fidelity is experimentally validated for the main variables of interest, as shown in Fig. 4; the validation procedure is described in Section IV. The model matches the measured pressure, flow, and displacement responses well, providing a reliable simulation platform for controller design and verification.

1) Influence of Pump Leakage Flow: From (6), the pump efficiency is mainly affected by the pump speed n_p , oil viscosity $\mu(T)$, and outlet pressure. Since n_p is prescribed, the focus is on $\mu(T)$ and pump pressure. The leakage flow increases with outlet pressure and decreases with $\mu(T)$. As HOT rises, $\mu(T)$ decreases and the BPV provides less throttling, which reduces back pressure and thus lowers the system pressure. These viscosity- and

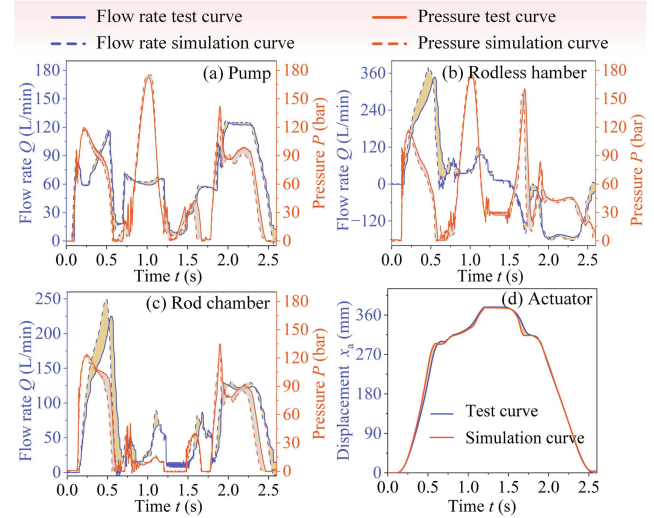


Fig. 4. Simulation-based validation of the SPCS model.

TABLE II
OIL TEMPERATURE-PUMP LEAKAGE COEFFICIENT

HOT (°C)	30	35	40	45	50
$C_{\text{leak}} (\times 10^{-3})$	2.7	3.8	5.0	6.3	8.0

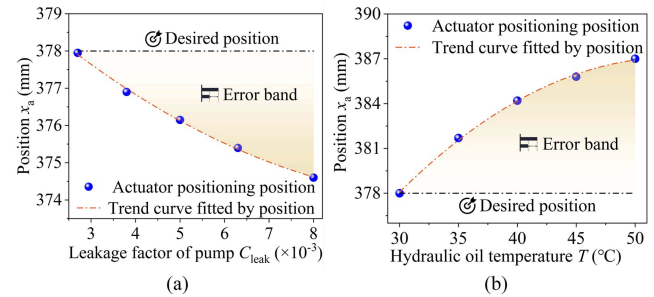


Fig. 5. Simulation scatter plots of actuator position under varying conditions. (a) Volumetric efficiency levels. (b) HOT.

pressure-dependent changes jointly reduce the pump volumetric efficiency.

Based on (6), leakage at different HOT is emulated by adjusting the pump leakage coefficient C_{leak} ; the HOT- C_{leak} mapping is given in Table II. These coefficients are applied in the simulation model for a commanded stroke of 378 mm. As shown in Fig. 5(a), the terminal position gradually shifts to about 374.5 mm. Moreover, as C_{leak} increases, the positioning error grows monotonically, indicating a pronounced degradation in motion accuracy.

2) Influence of System Back Pressure: Based on the SPCS and BPV models, actuator deceleration and final positioning are largely determined by the back pressure provided by the BPV. As HOT rises, viscosity $\mu(T)$ decreases, which weakens the throttling capacity of the damping orifices. Braking at the target is therefore reduced, leading to overshoot and a larger positioning error. Fig. 5(b) quantifies this effect. For a 378 mm stroke,

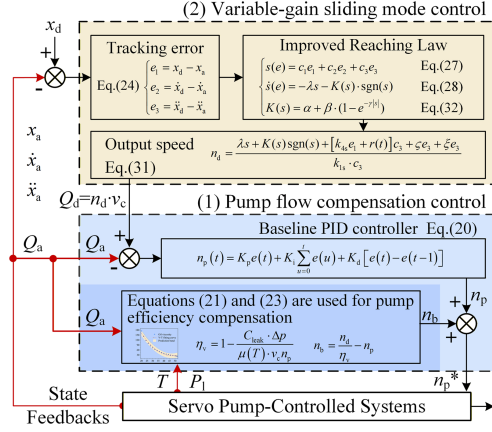


Fig. 6. Control block diagram of the SPCS.

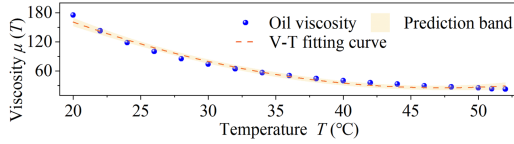


Fig. 7. Viscosity-temperature fitting curve of oil (ISO VG 46).

the terminal position shifts toward 387 mm as temperature rises, and the overshoot-induced error increases approximately linearly with temperature. Because pressure feedback control has finite delay, it is difficult to eliminate overshoot within the short deceleration interval of a high-inertia actuator. Targeted compensation for HOT-induced back-pressure degradation is thus required to improve temperature robustness.

HOT has a pronounced impact on both the driving and braking capability of SPCS. It can induce undershoot or overshoot, leading to positioning errors. These two error modes can couple, so the endpoint error varies from cycle to cycle. This variability degrades repeatability and ultimately reduces overall accuracy and stability.

III. CONTROL STRATEGY DESIGN

A. Control Architecture Design

Building on the baseline SPCS equipped with a position-based PID controller, a temperature-aware composite strategy that integrates pump-side internal compensation with a position-based VG-SMC. The goal is to mitigate HOT-induced degradation in positioning repeatability and accuracy. The control architecture is shown in Fig. 6. Each control cycle proceeds as follows:

- 1) *Step 1*: Given the position reference, define the tracking error and its derivatives as $E = [e_1, e_2, e_3]^T$.
- 2) *Step 2*: Based on VG-SMC, compute the corrected speed and flow demand associated with E .
- 3) *Step 3*: Compare the demanded flow with the measured pump flow, and apply primary closed-loop regulation via the baseline PID controller.

- 4) *Step 4*: In parallel path, estimate $\mu(T)$ from HOT measurements and apply the pump efficiency-based inversion amplification law to compensate temperature-induced flow nonlinearity.
- 5) *Step 5*: Synthesize the compensated command to obtain the final motor-speed reference n_p^* , apply it over the current sampling period, and feedback the updated state to Step 1.

B. Baseline PID Controller

PID control is a classical approach for SPCS. In the baseline system, the controller is driven by the discrete-time position-tracking error [37], and the control law is given by

$$n_p(t) = K_p e(t) + K_i \sum_{u=0}^t e(u) + K_d [e(t) - e(t-1)] \quad (20)$$

where $e(t)$ is the tracking error; K_p , K_i , and K_d are the PID gains; and $n_p(t)$ is the control output. The gains are tuned empirically for the target and kept constant during operation.

C. Pump Flow Compensation Control

An online viscosity-based channel is introduced to estimate the pump volumetric efficiency, as shown in Fig. 6(1). The objective is to compensate for the flow loss caused by efficiency degradation at elevated HOT. Based on the online estimate, the motor-speed command is increased accordingly, thereby restoring the effective supply flow.

Under ideal conditions, operating the pump at n_d would deliver the desired flow Q_d . However, internal leakage reduces the actual output flow to Q_a . Using (6), the pump volumetric efficiency is defined as

$$\eta_v = \frac{Q_a}{v_c \cdot n_p} = 1 - \frac{C_{leak} \cdot \Delta p}{\mu(T) \cdot v_c n_p}. \quad (21)$$

From (21), the pump flow-delivery capability is mainly determined by the n_p , $\mu(T)$, and Δp . Because $\mu(T)$ cannot be measured directly, the viscosity-temperature relation of the hydraulic oil offline is identified offline. The fitted $\mu(T)$ curve as shown in Fig. 7 and then embedded in the controller. As a result, $\mu(T)$ can be continuously estimated from the temperature sensor, and η_v can be computed in real-time using (21).

Based on the estimated η_v , the module corrects the pump command to offset leakage. Specifically, when the baseline PID requests a desired flow Q_d (equivalently n_d), the command is amplified so that the delivered flow Q_a tracks Q_d . In the ideal case, $Q_d = n_d \cdot v_c$, and the compensated command is obtained by scaling with $1/\eta_v$. The resulting compensation flow Q_b is

$$Q_b = Q_p^* - Q_a = \frac{Q_d}{\eta_v} - Q_a = \frac{Q_d \cdot \mu(T) \cdot v_c n_p}{\mu(T) \cdot v_c n_p - C_{leak} \cdot \Delta p} - Q_a \quad (22)$$

where Q_p^* is the compensated flow value; thus $Q_a = \eta_v \cdot Q_p^* = Q_d$.

Equivalently, the motor-speed command can be scaled by $1/\eta_v$, yielding the compensated speed n_b as

$$n_b = n_p^* - n_p = \frac{n_d}{\eta_v} - n_p \quad (23)$$

where n_p^* is the compensated motor (pump) speed command.

As shown in Fig. 6, this channel acts as a parallel feedforward path. The baseline command n_d , together with the measured T and Δp , is fed to the compensation module, which outputs the gain-corrected speed command n_b . When $\eta_v < 1$, n_b is increased to raise the speed and compensate the flow deficit.

D. Position-Based VG-SMC Control

To address the limited adaptability of the constant-gain PID under nonlinear conditions, a position-based VG-SMC is proposed to augment the baseline loop, as shown in Fig. 6(2).

First, the error vector using the actuator position-tracking error is defined and its derivatives

$$\begin{cases} e_1 = x_d - x_a \\ e_2 = \dot{x}_d - \dot{x}_a \\ e_3 = \ddot{x}_d - \ddot{x}_a \end{cases} \quad (24)$$

where x_d , $\dot{x}_d$, and $\ddot{x}_d$ are the desired position, velocity, and acceleration, respectively. Define e_1 , e_2 , and e_3 as the errors, respectively, so $\dot{e}_1 = e_2$, $\dot{e}_2 = e_3$, and $\dot{e}_3 = \ddot{x}_d - \ddot{x}_a$. Because the system is sampled at 0.002 s, higher order derivatives are computed using finite-difference approximations.

This article assumes that the reference trajectory x_d is piecewise C^3 . Its spectral bandwidth f_c satisfies $f_c \leq \kappa \cdot f_s$, where $\kappa \in [0.10.2]$ and f_s is the control sampling frequency. Under this assumption, x_d , $\dot{x}_d$, and $\ddot{x}_d$ can be stably estimated in discrete time using low-pass filtering and backward finite differences. The resulting estimation errors are lumped into the disturbance bound.

Linearizing the displacement-domain model in (19) about the operating point yields

$$\ddot{x}_a = k_{1s}n_d + k_{2s}\ddot{x}_a + k_{3s}\dot{x}_a + k_{4s}x_a + d(t) \quad (25)$$

where $k_{1s} = -\frac{A_e \beta_e v_c}{V_r m}$, $k_{2s} = -\frac{b}{m}$, $k_{3s} = -\frac{k}{m}$, $k_{4s} = \frac{A_e \cdot P_1 V_t}{V_r m}$. $d(t)$ is the lumped unmodeled disturbance of the system, assumed to satisfy $|d(t)| \leq \bar{d}$ for a known constant bound $\bar{d}$; and n_d denotes the desired pump speed

Substituting $\dot{e}_1$, $\dot{e}_2$, and $\dot{e}_3$ into (25) yields:

$$\dot{e}_3 = r(t) - k_{1s}n_d + k_{2s}e_3 + k_{3s}e_2 + k_{4s}e_1 - d(t). \quad (26)$$

Collect the terms that depend only on the known reference trajectory into the term: $r(t) = \ddot{x}_d - k_{2s}\ddot{x}_d - k_{3s}\dot{x}_d - k_{4s}x_d$.

For VG-SMC, the sliding surface is defined as a linear combination of the tracking errors

$$s(e) = C^T e = c_1 e_1 + c_2 e_2 + c_3 e_3 \quad (27)$$

where $C = [c_1 \ c_2 \ c_3]^T$ with $c_i > 0$ shapes the sliding-surface dynamics. Typically, c_i are chosen such that, on $s(e) = 0$, the position error e_1 converges with the desired dynamics.

In practice, the ideal sliding condition $s(t) \equiv 0$ is rarely achievable. Instead, quasi-sliding motion is characterized by convergence to a small boundary layer, $|s(t)| \leq \varepsilon$. Section III-E presents

the simulated $s(t)$ trajectory and shows that, during the dwell phase, it enters and stays within a bounded band around zero. This result supports the convergence and boundedness of the proposed sliding surface.

An improved reaching law is adopted. It includes a proportional term to reduce overshoot and a switching term to ensure fast convergence and disturbance rejection

$$\dot{s}(e) = -\lambda s - K(s) \cdot \text{sgn}(s) \quad (28)$$

where $\lambda > 0$ is the proportional decay coefficient and $K(s) > 0$ is the switching gain. The term $-\lambda s$ provides smooth exponential decay, whereas $-K(s) \cdot \text{sgn}(s)$ is discontinuous and maintains a nonzero reaching rate even when $|s|$ is small.

To enforce the reaching law, (27) is differentiated and substitute the error dynamics in (26). By setting $\dot{s}$ equal to the improved reaching law and simplifying, we obtain

$$\dot{s} = [k_{4s}e_1 + r(t) - d(t)] \cdot c_3 - k_{1s}n_d \cdot c_3 + \varsigma \cdot e_3 + \xi \cdot e_2 \quad (29)$$

where $\varsigma = c_2 + c_3 k_{2s}$, and $\xi = c_1 + c_3 k_{3s}$.

Substituting (29) into the reaching law and solving for the control input n_d yields

$$n_d = \frac{\lambda s + K(s) \text{sgn}(s) + [k_{4s}e_1 + r(t) - d(t)] c_3 + \varsigma e_3 + \xi e_2}{k_{1s} \cdot c_3} \quad (30)$$

where $\frac{k_{4s}e_1 + r(t)}{k_{1s}}$ is the model-based compensation term, $\frac{K(s) \text{sgn}(s)}{k_{1s} \cdot c_3}$ is the switching term, and $-\frac{d(t)}{k_{1s}}$ is an ideal disturbance cancellation. Since $d(t)$ is not measurable, the $d(t)$ -dependent term is omitted in implementation. Assuming $d(t)$ is bounded, its effect can be suppressed by choosing $K(s)$ large enough to cover the disturbance bound. The implemented control input is therefore:

$$n_d = \frac{\lambda s + K(s) \text{sgn}(s) + [k_{4s}e_1 + r(t)] c_3 + \varsigma e_3 + \xi e_2}{k_{1s} \cdot c_3}. \quad (31)$$

To prevent the smoothing term λs from degrading performance, a state-dependent variable switching gain $K(s)$ is employed, scheduled as follows:

$$K(s) = \alpha + \beta \cdot \left(1 - e^{-\gamma |s|}\right) \quad (32)$$

where $\alpha > 0$, $\beta > 0$, and $\gamma > 0$ are design constants. As $|s| \rightarrow 0$, α is the minimum switching gain, and $\alpha + \beta$ is the maximum. The parameter γ sets the transition rate from α to $\alpha + \beta$ with $|s|$. When $|s|$ is large, $K(s) \approx \alpha + \beta$ to accelerate convergence to the sliding surface. As $|s|$ decreases, $K(s) \rightarrow \alpha$, which reduces control effort and switching activity and thus alleviates chattering near $s = 0$. The boundedness $K(s) \in [\alpha, \alpha + \beta]$ is verified in Section III-E.

To verify that the control law in (31) drives the system to the sliding surface and maintains stability in the presence of bounded disturbances, the candidate Lyapunov function is

$$V = 0.5s^2. \quad (33)$$

Along the system trajectories, its time derivative is

$$\begin{aligned} \dot{V} &= s \dot{s} = s \left[-\lambda s - K(s) \text{sgn}(s) + \frac{d(t)}{k_{1s}} \right] \\ &= -\lambda s^2 - K(s) |s| + \frac{1}{k_{1s}} \cdot s \cdot d(t). \end{aligned} \quad (34)$$

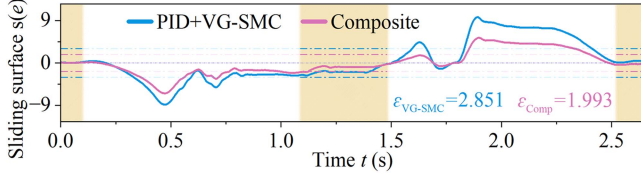


Fig. 8. Sliding surface $s(t) = C^T \cdot e$ for PID+VG-SMC and the composite controller.

Since $|s \cdot d(t)| \leq |s| \cdot |d(t)|$, the following upper bound as

$$\dot{V} \leq -\lambda s^2 - \left[K(s) - \frac{1}{k_{1s}} \cdot |d(t)| \right] \cdot |s|. \quad (35)$$

If the minimum gain α in (32) is chosen such that $\alpha \cdot k_{1s} \geq \bar{d}$, then $K(s) \cdot k_{1s} - |d(t)| \geq 0$ for all t . Hence, the $|s|$ term is nonpositive, and in the worst case we obtain

$$\dot{V} \leq -\lambda s^2 \leq 0. \quad (36)$$

Equation (36) guarantee $\dot{V} \leq 0$, so $V(t)$ is nonincreasing. By standard Lyapunov arguments, the sliding surface $s(e) = 0$ is reachable and invariant. On this surface, the error states (e_1, e_2, e_3) satisfy the reduced-order dynamics constrained by $c_1 \cdot e_1 + c_2 \cdot e_2 + c_3 \cdot e_3 = 0$. With c_1, c_2 , and c_3 chosen such that the corresponding linear differential equation is stable, it follows that $e_1(t) \rightarrow 0$ as $t \rightarrow \infty$, yielding asymptotic tracking. Therefore, the VG-SMC in (31) and (32) ensures convergence of the actuator position to the reference $x_d(t)$ in the presence of bounded external disturbances and modeling uncertainties.

E. Simulation Verification

To isolate the contribution of each control component before experimental testing, four configurations are evaluated: baseline PID controller, PID with pump-flow compensation (PFC), PID with VG-SMC, and the composite strategy integrating both PFC and VG-SMC. Simulations are performed under the commonly 378 mm stroke at HOT = 48°C to highlight high-temperature effects, including increased leakage and degraded back-pressure regulation.

First, Fig. 8 plots the sliding variable $s(t)$ during the dwell phase (shaded region) for PID+VG-SMC and the composite strategy, computed using (27). In both cases, $s(t)$ enters the quasi-sliding band and remains bounded. The boundary-layer widths are $\varepsilon_{VG-SMC} = \pm 2.851$ and $\varepsilon_{Comp} = \pm 1.993$. Compared with PID+VG-SMC, the composite strategy yields a narrower band and a smaller sliding residual, confirming the convergence and boundedness of the designed sliding surface.

Fig. 9(a) shows the scheduled switching gain $K(s)$ for PID+VG-SMC and the composite controller. Over the entire cycle, $K(s)$ remains bounded within the prescribed limits α and $\alpha + \beta$. During the dwell phase, as $|s| \rightarrow 0$, the gain decreases smoothly toward α , which reduces steady-state chattering and avoids excessive control effort.

Fig. 9(b) further presents the static mapping $K(|s|)$. It confirms the monotonic, state-dependent scheduling of the switching gain and the smooth transition governed by γ . Compared with

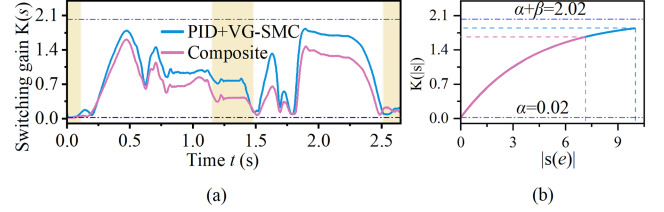


Fig. 9. Switching gain characteristics of PID+VG-SMC and the composite controller. (a) Scheduled switching gain $K(s)$; (b) Static mapping $K(|s|)$

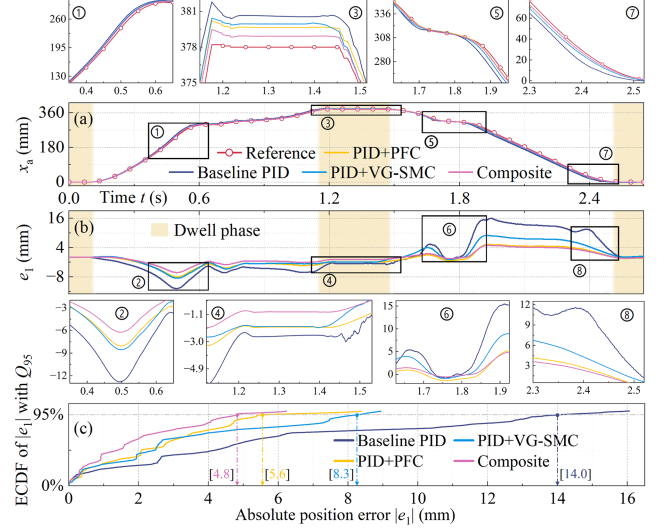


Fig. 10. Comparison of simulation tracking performance under four control strategies. (a) Actuator displacement responses; (b) Tracking errors; (c) ECDF of $|e_1|$.

PID+VG-SMC, the composite controller yields a lower $K(s)$ level during the dwell phase, which is consistent with its narrower sliding-residual band and reduced switching activity.

Fig. 10(a) and (b), together with the zoomed-in views, compare the actuator displacement responses and tracking errors under the four strategies. All controllers follow the reference at a macroscopic level. However, under high-HOT nonlinearities, the baseline PID exhibits the largest error fluctuations and the widest error band, most notably during the reversal and settling phases ((5)-(8)). In contrast, PID+PFC mainly corrects the systematic bias in the dwell segments by compensating flow loss via online pump-efficiency estimation, while PID+VG-SMC primarily suppresses residual uncertainties and dynamic disturbances during transients. By combining both mechanisms, the composite controller achieves the best tracking performance across all phases.

To characterize the error distribution, Fig. 10(c) shows the Empirical Cumulative Distribution Function (ECDF) of $|e_1|$, with the 95th percentile (Q_{95}) highlighted. Relative to baseline PID, the PID+PFC and composite curves shift left, indicating a higher probability of small errors; PID+VG-SMC also improves the distribution tail. Taken together, the time-domain results in Fig. 10(a) and (b) and the ECDF in Fig. 10(c) provide compact

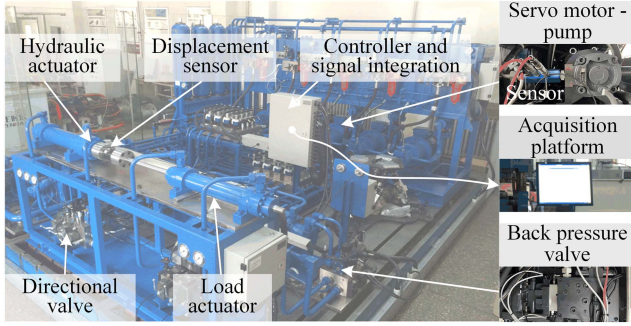


Fig. 11. Experimental test platform of the SPCS.

TABLE III
DETAILED EXPERIMENTAL OPERATING CONDITIONS

Case	1	2	3	4
Stroke (mm)	378	290	406	406
Velocity (%)	80%	70%	70%	99%
Purpose	Common	Short distance	Long distance	Speediness

TABLE IV
MINIMUM VALUE OF THE COST FUNCTION UNDER THE FOUR CONTROL STRATEGIES

Case	1	2	3	4
$J_{\text{Baseline}} (\text{mm}^2)$	7.61	2.32	3.75	18.33
$J_{\text{PID+PFC}} (\text{mm}^2)$	1.34	1.12	0.72	6.16
$J_{\text{PID+VG-SMC}} (\text{mm}^2)$	2.78	0.73	2.06	4.37
$J_{\text{Composite}} (\text{mm}^2)$	0.45	0.37	0.41	0.44

evidence that the proposed composite strategy achieves fast convergence with a tighter error bound under high-HOT nonlinearities, thereby setting clear expectations for the experiments in Section IV.

IV. EXPERIMENTS AND ANALYSES

A. Experimental Scenes Description

An SPCS test platform was built, as shown in Fig. 11. It integrates the functional units, sensors, signal modules, and the data-acquisition system. Several representative operating cases are considered. They include a common 378 mm stroke, a short-stroke case (290 mm), a long-stroke case (406 mm), and a high-speed case with the same stroke as the long-stroke condition. The test cases are parameterized by actuator stroke and velocity, which are selected according to the prescribed test ratios in the relevant standards [38]. The detailed settings are given in Table III.

Key system parameters are summarized in Table I. Controller parameters were first tuned manually and then refined by minimizing a cost function defined on the displacement tracking error. The baseline PID gains are $K_p = 260$, $K_i = 75$, and $K_d = 1.7$. The VG-SMC parameters are $c_1 = 60$, $c_2 = 10$, $c_3 = 0.73$, $\alpha = 0.02$, $\beta = 2$, and $\gamma = 30$. The minimum cost values for each control strategy are given in Table IV. The cost function is

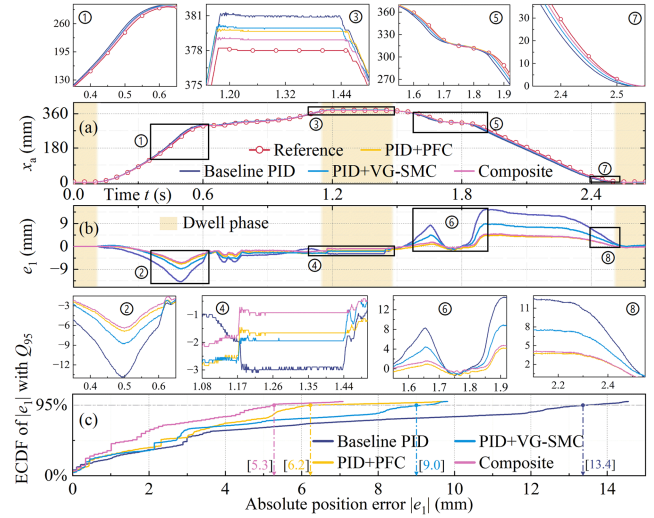


Fig. 12. Comparison of experimental tracking performance under four control strategies. (a) Actuator displacement responses; (b) Tracking errors; (c) ECDF of $|e_1|$.

defined as

$$J = \sum_{k=1}^N \frac{(x_a - x_d)^2}{N} \quad (37)$$

where x_a represents the actual system output at the k th time step, and x_d denotes the corresponding reference value.

B. Single-Cycle Control Response Analysis

To experimentally validate the simulation results in Section III, Fig. 12 shows the single-cycle displacement response under the Case 1 at HOT of 47°C–47.8°C. The overall behavior agrees well with the simulations: the baseline PID delivers the poorest tracking performance in the full-scale plot.

The benefit of PID+PFC is mainly observed in the dwell and quasi-dwell phases. The dwell offset is markedly reduced and the error band becomes tighter; for example, in zoom-in view ④ the holding error decreases from about 3 mm to 1 mm. In contrast, PID+VG-SMC primarily enhances transient robustness. Peak errors are significantly suppressed in the reversal phase: the 95th percentile of $|e_1|$ of drops by about 32.8%, and the reversal response becomes faster (see zoom-in view ③). More importantly, the composite controller combines both advantages, yielding the smallest error envelope across all phases while simultaneously reducing peak fluctuations and residual offset.

Fig. 12(c) summarizes the distribution of $|e_1|$. The error-threshold probabilities follow the same trend as in simulation. Notably, the experimental ECDF exhibits a slower rightward shift in the upper tail than the simulated one. This is expected and can be attributed to the combined effects of unmodeled dynamics (e.g., valve/pump hysteresis and sensor quantization) and test noise. Nevertheless, the performance ranking and the functional roles of the four strategies remain unchanged. Overall, the experiments corroborate the simulation-based interpretation: PFC

TABLE V
SYSTEM POSITIONING REPEATABILITY EVALUATION RESULTS

Case	1	2	3	4
Baseline PID controller (mm)	3.30	3.13	4.61	6.04
PID with VG-SMC (mm)	2.51	2.37	3.36	3.37
Optimization rate (%)	23.9	24.3	27.1	44.2
PID with flow-comp (mm)	2.36	2.46	3.15	3.43
Optimization rate (%)	28.5	21.4	31.7	43.2
Composite controller (mm)	2.06	2.29	2.89	3.08
Optimization rate (%)	37.6	26.8	37.3	49.0

TABLE VI
SYSTEM POSITIONING ACCURACY EVALUATION RESULTS

Case	1	2	3	4
Baseline PID controller (mm)	4.23	3.15	4.76	6.72
PID with VG-SMC (mm)	2.8	2.49	3.50	4.15
Optimization rate (%)	33.8	21.0	26.5	38.2
PID with flow-comp (mm)	2.1	2.74	3.31	4.28
Optimization rate (%)	50.4	13.0	30.5	36.3
Composite controller (mm)	0.92	2.27	3.12	1.91
Optimization rate (%)	78.3	27.9	34.5	71.6

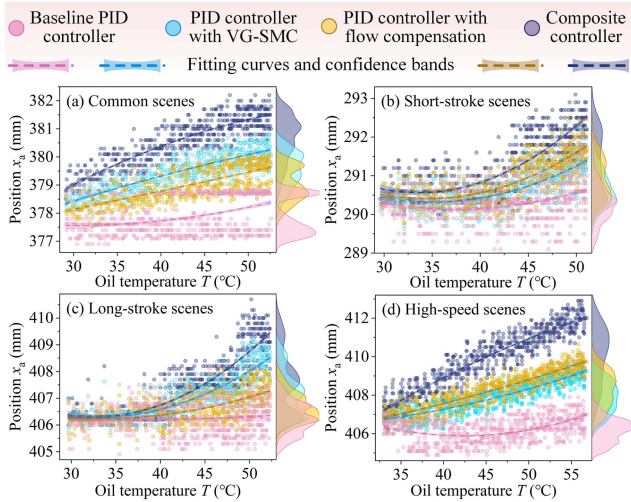


Fig. 13. Comparison of control optimization effects for SPCS.

mitigates source-side flow loss, VG-SMC improves transient robustness, and the composite controller yields the tightest error envelope.

C. Long-Cycle Performance Evaluation

After characterizing the intra-cycle time-domain behavior, this subsection further examines long-term positioning performance under hour-scale slowly varying temperature drift. For each case, more than 4000 motion cycles are executed continuously (over 4 h) to capture the gradual HOT drift and enable a comprehensive assessment of robustness and performance. For the four cases given in Table IV, long-term performance is quantified using the previously defined complementary metrics of positioning repeatability and positioning accuracy. The statistics are summarized in Table V and VI, confirming that the control configuration improvements in Fig. 13 consistently

translate into higher stability and accuracy over thousands of cycles under high-temperature conditions.

1) *Positioning Repeatability Evaluation*: Long-term positioning repeatability is evaluated using the weighted root-mean-square deviation defined in (38), which quantifies the cycle-to-cycle dispersion of the terminal position under slow HOT drift. Fig. 13 and Table V report the corresponding visualization and statistics. The baseline PID yields a repeatability metric of 3.13–6.04 mm. PID+PFC reduces it to 2.36–3.43 mm, corresponding to an improvement of about 21.4%–43.2% across all cases. PID+VG-SMC achieves a similar reduction to 2.37–3.37 mm, i.e., a 23.9%–44.2% improvement. The composite controller further tightens the metric to 2.06–3.08 mm, delivering the largest improvement of 26.8–49.0%. Notably, the largest gain occurs in the high-speed condition (CASE 4), where long-run dispersion is most sensitive to temperature-amplified nonlinearities and transient disturbances, thus best highlighting the benefit of the proposed “compensation + robustness” architecture

$$S_E = 4 \cdot \sqrt{\frac{1}{n-1} \cdot \sum_{i=1}^n (L_i - \bar{L})^2} \quad (38)$$

where S_E is the repeatability index; L_i is the terminal position at the i -th cycle; $\bar{L}$ is the mean terminal position; and n is the number of cycles.

2) *Positioning Accuracy Evaluation*: To evaluate worst-case terminal positioning accuracy under high-temperature drift, the maximum absolute error metric is defined in (39). Fig. 13 and Table VI show that, relative to the baseline PID (3.15–6.72 mm), PID+PFC constrains the maximum terminal error to 2.10–4.28 mm, corresponding to improvements from 13.0% (case 2) to 50.4% (case 1). This indicates that its benefit is condition dependent when the dominant error source shifts between supply-flow deficit bias and disturbance-driven fluctuations. PID+VG-SMC yields a maximum terminal error of 2.49–4.15 mm, providing a consistent improvement of 21.0%–38.2% across all cases. More importantly, the composite controller further reduces the maximum terminal error to 0.92–3.12 mm and achieves improvements exceeding 70% in cases 1 and 4, directly demonstrating its strong capability to suppress HOT-induced positioning errors over thousands of cycles

$$S_A = \text{Max} \{|L_i - L_s|\} \quad (39)$$

where S_A represents the positioning accuracy evaluation value, and L_s refers to the expected set motion position.

V. CONCLUSION

This article investigates the mechanisms responsible for performance degradation in SPCS under HOT rise during long-duration operation. It then proposes and validates a temperature-tolerant composite control framework that integrates a temperature-aware pump-side inner compensation loop with a position-domain VG-SMC robust outer loop. Specifically, an inverse-amplification scheme based on online pump-efficiency estimation compensates the HOT-induced loss in flow

delivery, while the outer-loop VG-SMC suppresses multisource uncertainties and dynamic disturbances. This architecture yields decoupled yet synergistic benefits in systematic bias reduction and peak-fluctuation suppression. Single-cycle and long-cycle experiments show that, compared with the baseline PID, the proposed strategy improves both repeatability and positioning accuracy by more than 25%, and achieves accuracy gains exceeding 70% under nominal and high-speed conditions. These results confirm strong robustness to HOT disturbances and excellent temperature tolerance. Despite these gains, practical limitations remain. Under high measurement noise, limited bandwidth, or insufficient quantization resolution, differential operations may amplify high-frequency components, leading to chattering and residual steady-state bias. Mitigating these effects will be the focus of future work.

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